Data collection

5 public tissue cohorts

TCGA-BRCA	PRJNA294226	PRJNA482141	PRJNA808405	PRJNA934049
tissue	tissue	tissue	tissue	tissue
n = 194	*file pending*	*file pending*	*file pending*	*file pending*
187 T / 7 N				

2 in-house plasma cohorts

yyfbatch1	yyfbatch2
plasma	plasma
n = 192	n = 212
106 T / 86 N	119 T / 93 N

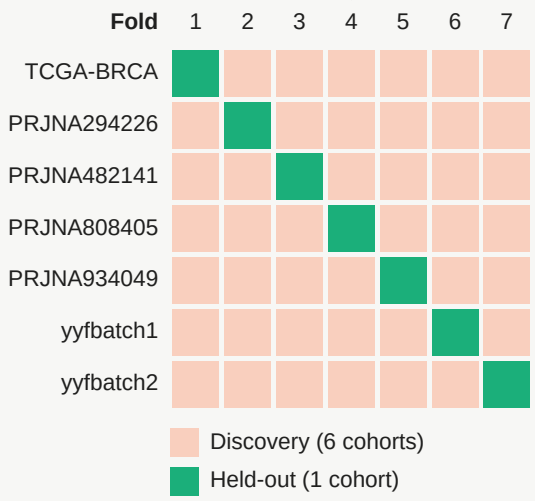
Preprocessing

Harmonise labels — Tumor vs Normal → **Shared piRNAs** — present in every cohort → **log2(TPM + 1)** — transformation → **Per-cohort z-score** — label-free, within each cohort

Leave-one-cohort-out validation

7 folds · in each fold one cohort is held out as the independent validation set; the other 6 form the discovery set

next fold — repeat until every cohort has been held out once



Step 1 *discovery cohorts only*

Feature screening

1. **Expression filter** — detected in ≥ 50% of samples of every discovery cohort
2. **Differential expression** — Welch t-test in each cohort, Tumor vs Normal → signed z
3. **Direction consistency** — same Tumor/Normal direction in all 6 discovery cohorts
4. **Meta-ranking** — Stouffer combined |z| → top 10 candidate piRNAs

Step 2 *locked before validation*

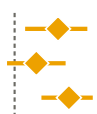
Model building & verification

1. **Logistic regression** — L2-penalised, class-balanced, pooled discovery cohorts
2. **Internal estimate** — 5-fold CV in discovery data, Step 1 re-run inside each fold
3. **Lock model + threshold** — Youden's J on discovery data
4. **Independent validation** — held-out cohort: AUC (DeLong 95% CI), sensitivity, specificity

held-out cohort → independent validation only

Across the 7 folds

Performance — discovery-CV vs held-out AUC per fold (ROC, forest plot)



Selection stability — how often each piRNA is selected across the folds

Final signature — Step 1 and model fit re-run on all 7 cohorts → final model

Functional prediction of the signature piRNAs

TCGA-BRCA mRNA-seq — n = 23 samples matched to TCGA piRNA data, 16,330 genes → **piRNA–mRNA correlation** — Pearson r for every gene detected in ≥ 70% of samples, FDR < 0.05, |r| > 0.3 → **Pathway enrichment** — GO (BP, CC, MF), KEGG, Reactome over-representation, GSEA on genes ranked by r → **Functional network** — piRNA – gene – pathway

